

Uncertainty Quantification for Out-of-Distribution Detection: A Comparative Study with a Focus on Conformal Prediction

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Abstract

Reliable confidence estimates are essential for deploying deep neural networks in safety-critical or open-world settings, where inputs may come from a distribution the model has never seen. This paper compares four uncertainty quantification (UQ) methods – softmax confidence, Monte Carlo Dropout, deep ensembles, and split conformal prediction with adaptive prediction sets (APS) – on the task of out-of-distribution (OOD) detection and confidence calibration. Using CIFAR-10 as the in-distribution (ID) dataset and SVHN as the OOD dataset, we evaluate OOD detection and robustness under simulated distribution shift for all methods, and we report calibration results for the softmax baseline, with a full calibration comparison left as a limitation and future work. Our results show that deep ensembles substantially outperform softmax confidence and MC-Dropout for OOD detection (AU-ROC 0.829 vs. 0.720 and 0.715, respectively), and that conformal prediction sets grow appropriately with input difficulty, offering a favorable balance between statistical guarantees and computational cost. We discuss the practical trade-offs between these methods and their implications for trustworthy ML deployment.

1 Introduction

Deep neural networks are increasingly deployed in settings where they may encounter inputs that differ substantially from their training distribution. A classifier that is highly confident on such out-of-distribution (OOD) inputs – while being wrong – poses a serious risk in safety-critical applications such as autonomous driving, medical diagnosis, and industrial monitoring. This motivates the need for models that not only predict well, but also *know when they do not know*.

Uncertainty quantification (UQ) methods aim to equip predictive models with calibrated confidence estimates that reflect the true likelihood of correctness, and that degrade gracefully – rather than silently – when inputs move outside the training distribution.

Research question. How well do different UQ methods detect out-of-distribution inputs and produce reliable confidence estimates, with a particular focus on conformal prediction?

Contributions.

- An empirical comparison of four UQ methods – softmax confidence, MC-Dropout, deep ensembles, and split conformal prediction (APS) – under a common evaluation protocol.
- An OOD detection benchmark using CIFAR-10 (ID) and SVHN (OOD), together with a calibration analysis of the softmax baseline.
- A discussion of the practical compute/accuracy/guarantee trade-offs relevant to deploying UQ methods in practice.

The remainder of this paper is organized as follows. Section 2 reviews related work on uncertainty estimation and conformal prediction. Section 3 describes the datasets, base model, and each UQ method. Section 4 details the experimental setup. Section 5 presents results, Section 6 discusses failure cases and trade-offs, and Section 7 concludes.

2 Background and Related Work

2.1 Aleatoric and Epistemic Uncertainty

Predictive uncertainty is commonly decomposed into *aleatoric* uncertainty, arising from inherent noise in the data, and *epistemic* uncertainty, arising from limited knowledge of the model (e.g. due to insufficient training data in a region of input space) [4]. OOD detection is primarily concerned with epistemic uncertainty: the model should express high uncertainty on inputs unlike anything seen during training.

2.2 Softmax Confidence

The maximum softmax probability is the simplest confidence signal available from any classifier, but it is well known to be poorly calibrated and prone to overconfidence, including on inputs far from the training distribution [2, 3].

2.3 Monte Carlo Dropout

MC-Dropout approximates Bayesian inference by keeping dropout active at test time and performing multiple stochastic forward passes, treating the resulting prediction variance as an epistemic uncertainty estimate [1].

2.4 Deep Ensembles

Deep ensembles train several independently initialized models and use the disagreement between their predictions as an uncertainty signal. Despite their simplicity, ensembles are widely reported as a strong, well-calibrated baseline for uncertainty estimation [6].

2.5 Conformal Prediction

Conformal prediction provides distribution-free, finite-sample coverage guarantees: given a desired miscoverage rate α , it produces prediction sets that contain the true label with probability at least $1 - \alpha$, under only an exchangeability assumption [9]. Adaptive Prediction Sets (APS) improve on simple threshold-based conformal scores by using the cumulative sum of sorted softmax probabilities, which tends to produce more balanced coverage across classes and produces larger sets for harder or more ambiguous inputs [8].

3 Methodology

3.1 Datasets

We use CIFAR-10 [5] as the in-distribution (ID) dataset: 10 classes of 32×32 RGB natural images, with a standard 50,000/10,000 train/test split. SVHN [7], a dataset of 32×32 RGB house-number images with disjoint label semantics, is used as the out-of-distribution (OOD) dataset; we draw OOD evaluation examples from its test split. Because CIFAR-10 and SVHN share the same input resolution and channel count, all methods can be applied to both datasets without architectural changes, isolating the comparison to differences in the input distribution itself rather than confounding it with input shape.

The base classifier is trained only on CIFAR-10; SVHN images are never seen during training or model selection, so they serve as a genuinely out-of-distribution test set for every UQ method. Of the CIFAR-10 test set, 20% is held out as a calibration split for the conformal prediction procedure (Section 3.4), with the remaining 80% used to evaluate ID accuracy, calibration, and OOD detection for all four methods.

3.2 Base Model

A small convolutional neural network is used as the base classifier. The model architecture is deliberately kept small: the UQ analysis is the contribution of this work, not the base model’s predictive performance, and a small model keeps training tractable on limited compute.

3.3 Uncertainty Quantification Methods

3.3.1 Softmax Confidence

For an input x and logits $z(x) \in \mathbb{R}^K$ over K classes, the predicted class and its confidence are

$$p(y = c | x) = \frac{\exp(z_c(x))}{\sum_{k=1}^K \exp(z_k(x))}, \quad (1)$$

$$\hat{y} = \arg \max_c p(y = c | x), \quad (2)$$

$$\text{conf}(x) = \max_c p(y = c | x), \quad (3)$$

taken directly from the trained classifier with no additional computation.

3.3.2 MC-Dropout

Dropout (rate $p = 0.3$) is kept active at inference time. For each input x , we perform $T = 30$ stochastic forward passes $\{p_t(y | x)\}_{t=1}^T$ and use the mean predictive distribution for classification:

$$\bar{p}(y = c | x) = \frac{1}{T} \sum_{t=1}^T p_t(y = c | x). \quad (4)$$

The predictive uncertainty is taken as the predictive entropy of the averaged distribution,

$$\mathcal{H}(x) = - \sum_{c=1}^K \bar{p}(y = c | x) \log \bar{p}(y = c | x), \quad (5)$$

or, alternatively, the variance of the T stochastic predictions for the predicted class,

$$\sigma^2(x) = \frac{1}{T} \sum_{t=1}^T (p_t(\hat{y} | x) - \bar{p}(\hat{y} | x))^2. \quad (6)$$

3.3.3 Deep Ensembles

We train $M = 5$ independently initialized copies of the base model. Writing $p_m(y | x)$ for the softmax output of ensemble member m , the ensemble prediction is the average across members,

$$\bar{p}(y = c | x) = \frac{1}{M} \sum_{m=1}^M p_m(y = c | x), \quad (7)$$

and disagreement between members – measured analogously to Eq. (5)–(6) with M in place of T – is used as the uncertainty score.

3.3.4 Split Conformal Prediction (APS)

Let $\pi_1(x), \pi_2(x), \dots, \pi_K(x)$ denote the class indices sorted so that $p(y = \pi_1 | x) \geq p(y = \pi_2 | x) \geq \dots$. For a calibration example (x_i, y_i) with true label at sorted rank $o(x_i, y_i)$, the APS nonconformity score with randomization $u_i \sim \text{Uniform}(0, 1)$ is

$$s(x_i, y_i) = \sum_{j=1}^{o(x_i, y_i)-1} p(y = \pi_j | x_i) + u_i \cdot p(y = \pi_{o(x_i, y_i)} | x_i). \quad (8)$$

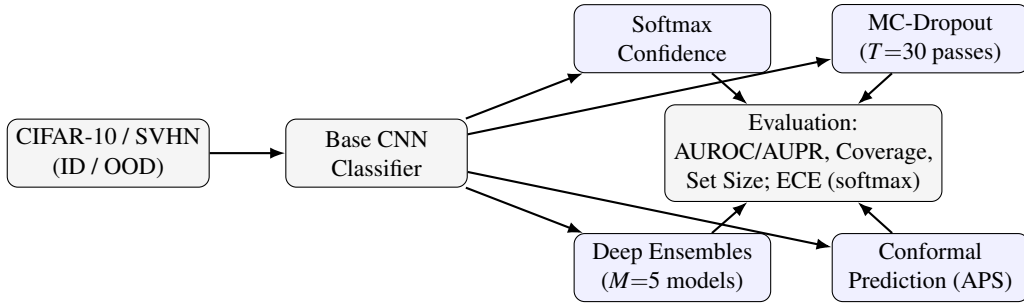


Figure 1: Overall methodology pipeline. A single base CNN classifier is trained on CIFAR-10 and reused across all four uncertainty quantification methods, each producing an uncertainty score that is compared under a common evaluation protocol on ID and OOD inputs.

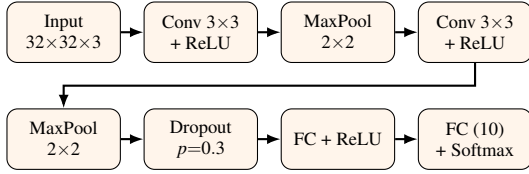


Figure 2: Base CNN classifier architecture. The dropout layer is kept active at test time for the MC-Dropout method (Section 3), while the same architecture is trained independently five times for the deep ensemble.

Given n calibration points, the threshold is the empirical $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $\{s(x_i, y_i)\}_{i=1}^n$:

$$\hat{q} = \text{Quantile} \left(\{s(x_i, y_i)\}_{i=1}^n; \frac{\lceil (n+1)(1-\alpha) \rceil}{n} \right). \quad (9)$$

At test time, the prediction set for a new input x is

$$C(x) = \left\{ \pi_j(x) : \sum_{k=1}^{j-1} p(y=\pi_k | x) + u \cdot p(y=\pi_j | x) \leq \hat{q} \right\}, \quad (10)$$

with $\alpha = 0.1$, targeting 90% marginal coverage $\Pr(y \in C(x)) \geq 1 - \alpha$. The randomization term u in Eqs. (8)–(10) is essential: omitting it biases $\hat{q}$ toward 1.0 and yields meaningless coverage/set-size numbers [8].

3.4 Evaluation Metrics

- **Calibration:** reliability diagrams and Expected Calibration Error (ECE). Predictions are partitioned into B equal-width confidence bins $\{b_1, \dots, b_B\}$, and ECE is the confidence-weighted average gap between accuracy and confidence within each bin:

$$\text{ECE} = \sum_{b=1}^B \frac{|b|}{n} |\text{acc}(b) - \text{conf}(b)|, \quad (11)$$

where $|b|$ is the number of samples in bin b , n the total number of samples, $\text{acc}(b)$ the fraction of correct predictions in b , and $\text{conf}(b)$ the average predicted confidence in b .

Table 1: Experimental configuration.

Parameter	Value
ID dataset	CIFAR-10
OOD dataset	SVHN
Batch size	128
Epochs	15
Learning rate	1×10^{-3}
Ensemble members	5
MC-Dropout passes	30
MC-Dropout rate	0.3
Conformal α	0.1 (target 90% coverage)
Calibration fraction	0.2
Noise severities (stress test)	0.0, 0.05, 0.1, 0.2, 0.35, 0.5

- **OOD detection:** AUROC and AUPR, treating “is this sample OOD” as a binary classification problem using each method’s uncertainty score (Eqs. (3), (5)) as the decision score.
- **Conformal coverage:** empirical coverage on the ID test set, $\frac{1}{n} \sum_{i=1}^n \mathbf{1}\{y_i \in C(x_i)\}$, compared against the target $1 - \alpha$.
- **Set size:** average conformal prediction set size $\frac{1}{n} \sum_{i=1}^n |C(x_i)|$, on both ID and OOD inputs.

4 Experimental Setup

Table 1 summarizes the experimental configuration. All experiments use a fixed random seed for reproducibility.

To evaluate robustness under distribution shift, we simulate corruption by injecting Gaussian noise of increasing standard deviation directly into the ID test inputs, rather than relying on the separately-hosted CIFAR-10-C dataset. This keeps the pipeline self-contained while still exercising each method’s behavior as inputs move progressively further from the training distribution.

Table 2: OOD detection performance on CIFAR-10 (ID) vs. SVHN (OOD).

Method	AUROC	AUPR
Softmax	0.720	0.588
MC-Dropout	0.715	0.569
Deep Ensembles	0.829	0.750

5 Results

5.1 Comparative Analysis

Table 2 summarizes OOD detection performance (AUROC/AUPR) for the three confidence-score-based methods, and Table 3 reports the conformal prediction results separately, since coverage and set size are not directly comparable to an AUROC/AUPR score. Deep ensembles achieve the strongest OOD detector by a clear margin, with AUROC 0.829 versus 0.720 for softmax and 0.715 for MC-Dropout – roughly 11 points above the next-best method (Figure 3). MC-Dropout does not improve on the plain softmax baseline in our setup; with $T = 30$ stochastic passes on a comparatively small CNN, the dropout masks appear not to inject enough predictive diversity to separate ID from OOD inputs any better than the single deterministic forward pass already does. This is consistent with reports that MC-Dropout’s benefit is architecture- and capacity-dependent [1], rather than a universal improvement over softmax confidence.

For calibration, the softmax baseline has $ECE = 0.0461$ on the ID test set, with the reliability diagram showing systematic *overconfidence* in the high-confidence bins (predicted confidence exceeds observed accuracy above ~ 0.6). We did not compute ECE for MC-Dropout or the deep ensemble in this run; extending the calibration comparison to all three confidence-based methods is left as immediate follow-up work (Section 6).

The split conformal predictor (APS) achieves empirical ID coverage of 0.9525 against a target of $1 - \alpha = 0.90$ (Table 3), with an average prediction set size of 1.95 out of 10 classes – small enough to be practically useful while still exceeding the coverage guarantee. On OOD (SVHN) inputs, the average set size grows to 2.64, showing that the method appropriately expresses more uncertainty when it is fed inputs the base classifier was never trained to classify, even though conformal prediction is not designed as an OOD detector per se. The empirical coverage is somewhat above the 90% target; this is expected given that our implementation randomizes the calibration nonconformity score (Eq. (8)) but constructs test-time prediction sets deterministically rather than also randomizing inclusion of the boundary class, which is a common, slightly conservative simplification of the full APS procedure [8].

5.2 Robustness Under Distribution Shift

We simulate distribution shift by injecting Gaussian noise of increasing standard deviation σ directly into ID test inputs

Table 3: Split conformal prediction (APS) results, $\alpha = 0.1$.

Quantity	Value
Calibrated threshold $\hat{q}$	0.9413
ID coverage (target 0.90)	0.9525
Avg. set size, ID	1.95
Avg. set size, OOD	2.64

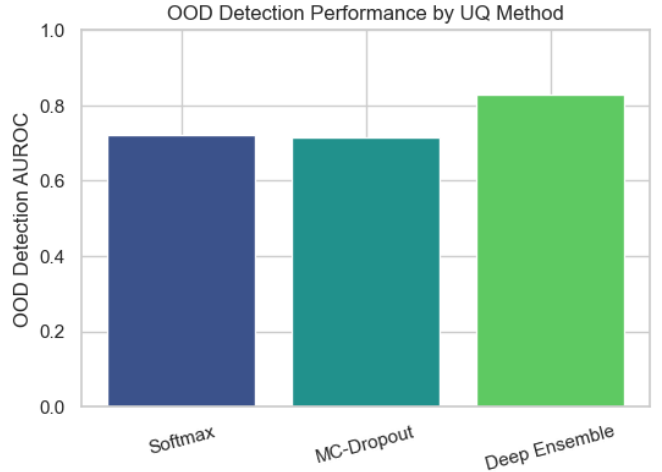


Figure 3: OOD detection AUROC by uncertainty quantification method. Deep ensembles outperform both softmax confidence and MC-Dropout.

(Section 4) and re-evaluate the base classifier’s accuracy and confidence, together with the conformal predictor’s coverage and set size, at each severity level (Figure 4). Accuracy degrades sharply and roughly monotonically with σ , from 0.801 at $\sigma = 0$ to 0.192 at $\sigma = 0.5$ – an 8-fold drop – while mean softmax confidence falls much more slowly, from 0.847 to 0.640. This growing gap between accuracy and confidence is exactly the overconfidence-under-shift failure mode that motivates uncertainty quantification: by $\sigma = 0.5$ the model is on average still 64% confident while it is correct only 19% of the time.

Conformal coverage tracks this same degradation: it starts above the 90% target (0.9525 at $\sigma = 0$) but falls monotonically as noise increases, reaching 0.535 at $\sigma = 0.5$ – well below the nominal guarantee. This is expected rather than a failure of the method: split conformal’s marginal coverage guarantee relies on the calibration and test data being exchangeable, an assumption that Gaussian corruption directly violates once σ grows large enough to shift the input distribution. Average prediction set size compensates partially, growing from 1.95 classes at $\sigma = 0$ to 3.73 at $\sigma = 0.5$ as the method’s per-input uncertainty signal increases with corruption, but this growth is not enough to preserve the coverage guarantee under this exchangeability violation – a limitation of vanilla split conformal prediction that motivates distribution-shift-aware variants in the literature.

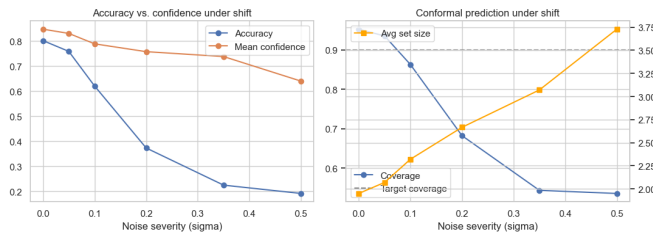


Figure 4: Left: base classifier accuracy and mean softmax confidence as a function of injected Gaussian noise severity σ ; the growing gap between the two curves is the overconfidence-under-shift failure mode. Right: conformal coverage (target $1 - \alpha = 0.90$, dashed line) and average prediction set size under the same shift. Coverage degrades once shift becomes severe enough to break the exchangeability assumption, while set size grows only partially in compensation.

6 Discussion

6.1 Error Analysis

We qualitatively examine failure cases identified in the accompanying experimental notebook (Section 8). (i) The softmax baseline’s most confidently wrong predictions are concentrated among visually similar class pairs – e.g., *airplane* misclassified as *ship*, *automobile* as *truck*, and *bird* as *horse* – each with confidence 1.000, illustrating that high softmax confidence offers no protection against errors on ambiguous within-distribution images. (ii) Among OOD (SVHN) inputs, the conformal predictor’s worst-case failures to flag genuine OOD samples produced prediction sets of size 1 – the smallest possible non-empty set – showing that even a distribution-free method can occasionally be misled into near-certainty on inputs it was never trained to classify.

6.2 Compute Trade-offs

The four methods differ substantially in computational cost: softmax confidence is free (a single forward pass); MC-Dropout requires $T = 30\times$ the inference cost of a single pass; deep ensembles require $M = 5\times$ both training and inference cost; and conformal prediction adds a near-negligible calibration step on top of any base classifier. This makes conformal prediction particularly attractive when a distribution-free coverage *guarantee* is needed without the overhead of ensembling or repeated sampling.

6.3 Limitations

This study uses a single random seed per configuration, a single ID/OOD dataset pair (CIFAR-10 vs. SVHN), and simulated rather than natural corruption for the robustness analysis. These choices keep the pipeline tractable but limit the generality of the conclusions.

Calibration was only evaluated for the softmax baseline; extending ECE to MC-Dropout and deep ensembles is left as future work. This limits the scope of our calibration comparison

but does not affect the OOD detection and conformal coverage results.

7 Conclusion and Future Work

This study compared four uncertainty quantification methods on CIFAR-10 (ID) vs. SVHN (OOD). Deep ensembles were the clear best OOD detector (AUROC 0.829), while MC-Dropout offered no improvement over plain softmax confidence in this setup, consistent with prior reports that its benefit depends on model architecture and capacity. Split conformal prediction met its 90% coverage target on clean ID data (95.25% empirical coverage, average set size 1.95) and expanded its sets on OOD inputs (2.64), but its coverage guarantee degraded sharply under simulated distribution shift – falling to 53.5% at the highest noise severity – because Gaussian corruption violates the exchangeability assumption the guarantee relies on. Taken together, these results suggest deep ensembles are preferable when compute allows, while conformal prediction is attractive for its near-zero overhead and formal guarantees on unshifted data, but should not be trusted to maintain coverage under distribution shift without shift-aware extensions. Future work includes evaluating on natural corruptions (CIFAR-10-C), additional OOD dataset pairs, larger backbone architectures, and extending conformal risk control beyond simple classification accuracy.

Data Availability Statement

This study uses two publicly available benchmark datasets. CIFAR-10 [5] is available at <https://www.cs.toronto.edu/~kriz/cifar.html>. SVHN [7] is available at <http://ufldl.stanford.edu/housenumbers/>. No new data were collected for this work, and no restrictions apply to accessing either dataset.

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